Powers of Ten and Place-Value Relationships

Answer Key

Grades 4–5   Number and Operations in Base Ten

Quick Answers

1. 10	4. 0.1	7. 3600	10. 45
2. 1000	5. 8000	8. 100	11. 30705
3. 4700	6. 60000	9. 3300	12. 203

Common wrong answers

5. *If they got 8100: rounded to the nearest hundred instead of the nearest thousand*
8. *If they got 10: answered 10 because one place left is always 10 times, ignoring that the digits sit two places apart*
9. *If they got 3200: rounded 2905 up to 3000 by looking at the tens digit instead of the tens place value*
10. *If they got 450000: multiplied by 100 instead of dividing, shifting the digits the wrong way*
11. *If they got 375: ignored the empty thousands and tens places and wrote the digits side by side as 375*
12. *If they got 270.67: rounded the divisor down to 30 instead of to the nearest ten*

1. Two 4s in 4400.

Step 1. "How many times as much" is a comparison of two *values*, not two digits, so write each digit's value first: thousands 4 $\to$ 4000, hundreds 4 $\to$ 400.
Step 2. Compare by dividing the larger value by the smaller: $4000 \div 400 = 10$. $\boxed{10}$
Why: thousands and hundreds are neighbouring places, and every place is worth ten times the place to its right.

2. The meaning of 10^3.

Step 1. The exponent counts factors of ten, so $10^3 = 10 \times 10 \times 10$.
Step 2. Multiply: $10 \times 10 = 100$, and $100 \times 10 = 1000$. $\boxed{1000}$
Pattern to notice: the exponent also counts the zeros — 10^3 has three zeros, and it names the thousands place.

3. 47×100.

Step 1. $100 = 10^2$, so this multiplication makes every digit worth 10^2 times as much — a shift of *two places to the left.*
Step 2. The 4 moves from tens to thousands and the 7 moves from ones to hundreds; two empty places are filled with zeros as placeholders: $47 \to 4700$. $\boxed{4700}$
Language that matters: the digits move, the decimal point does not "gain zeros". Zeros appear because the ones and tens places are now empty.

4. Two 8s in 0.88.
Step 1. Write each digit's value: tenths $8 \to 0.8$, hundredths $8 \to 0.08$. The question compares the smaller with the larger, so divide in that order.
Step 2. $0.08 \div 0.8 = 0.1$, which is one tenth. $\boxed{0.1}$
Why: moving one place to the *right* divides a digit's value by ten. This is the same relationship as problem 1, read in the other direction.

5. Estimate $4780 + 3260$ to the nearest thousand.
Step 1. Rounding to the nearest 10^3 means looking at the hundreds digit of each number: 4780 has 7 hundreds, so it rounds up to 5000; 3260 has 2 hundreds, so it rounds down to 3000.
Step 2. Add the rounded numbers: $5000 + 3000 = 8000$. $\boxed{8000}$
Common wrong answer: 8100 — rounded to the nearest hundred $(4800 + 3300)$ instead of the nearest thousand.

6. 60000 as a digit times a power of ten.
Step 1. The leading digit is 6 and it sits in the ten-thousands place, and ten-thousands is 10^4, so the exponent is 4: $60000 = 6 \times 10^4$.
Step 2. Check by multiplying back: $6 \times 10^4 = 6 \times 10000 = 60000$. $\boxed{60000}$
Check that transfers: count the zeros. Four zeros after the 6 means 10^4, every time.

7. 3.6×1000.
Step 1. $1000 = 10^3$, so every digit shifts three places to the left.
Step 2. The 3 moves from ones to thousands and the 6 moves from tenths to hundreds. The tens and ones places are now empty, so they take zeros: $3.6 \to 3600$. $\boxed{3600}$
Watch for: students who "add three zeros" write 3.6000. Shifting digits is the reliable description; adding zeros is not.

8. Two 5s in 55500, two places apart.
Step 1. Values first: ten-thousands $5 \to 50000$, hundreds $5 \to 500$. The places are ten-thousands, thousands, hundreds — so the digits are *two* places apart, not one.
Step 2. Two places means two factors of ten: $50000 \div 500 = 100 = 10^2$. $\boxed{100}$
Common wrong answer: 10 — the "one place left is ten times" rule applied without counting how many places apart the digits actually are.

9. Estimate $6238 - 2905$ to the nearest hundred.
Step 1. Rounding to the nearest 10^2 means looking at the tens digit: 6238 has 3 tens, so it rounds down to 6200; 2905 has 0 tens, so it rounds down to 2900.
Step 2. Subtract the rounded numbers: $6200 - 2900 = 3300$. $\boxed{3300}$
Common wrong answer: 3200 — 2905 rounded up to 3000 because the 5 in the ones place was mistaken for a tens digit. Round by the place you were asked for, and read the digit to its right.

10. $4500 \div 100$.
Step 1. $100 = 10^2$, and dividing makes each digit worth one hundredth as much, so the digits shift *two places to the right.*
Step 2. The 4 moves from thousands to tens and the 5 moves from hundreds to ones: $4500 \to 45$. $\boxed{45}$
Common wrong answer: 450000 — multiplied by 100 instead of dividing. A quotient smaller than the dividend is the check.

11. Expanded form with powers of ten.

(a) Term by term: $3 \times 10^4 = 30000, \quad 7 \times 10^2 = 700, \quad$ and the last term is $5 \times 10^0 = 5$.

(b) Add them, keeping each digit in its own place: $30000 + 700 + 5$. There is no thousands term and no tens term, so those places are empty and take 0 as a placeholder. $\boxed{30705}$

Common wrong answer: 375 — the digits written side by side with the empty places dropped. Placeholders are what keep every other digit in its place.

12. Estimate $8120 \div 39$.

(a) 39 rounds to 40, because 9 ones is more than half of ten. The useful part is that $40 = 4 \times 10$, so the division becomes a fact plus a shift.

(b) $8120 \div 40$: divide by 10 first (shift one place right) to get 812, then divide by 4 to get 203. $\boxed{203}$

(c) The exact quotient is about 208, so the estimate is *smaller* than the exact answer. Rounding the divisor *up* means sharing into bigger groups, and bigger groups give a smaller quotient. (Accept any answer that says the estimate is smaller and links it to the larger divisor.)

Common wrong answer: about 271 — 39 rounded down to 30 rather than to the nearest ten.